EXPERIMENT 6:

AIM:

i) To generate a Phase Modulated (PM) signal in MATLAB using specified carrier and message signal parameters.

ii) To compute the frequency spectrum of the PM signal using the Fast Fourier Transform (FFT).

iii) To calculate and plot the Power Spectral Density (PSD) of the PM signal using MATLAB.

iv) To identify the carrier component and sideband frequencies present in the PM spectrum.

v) To analyze the effect of phase modulation on the spectral distribution and bandwidth of the transmitted signal.

vi)To investigate the influence of modulation index on the Power Spectral Density characteristics of the PM signal.

CODE:

```
clc; clear; close all;

Am=1; fm=100; Ac=5; fc=1000;
fs=20*fc; kp=pi/2;
t=0:1/fs:4/fm;

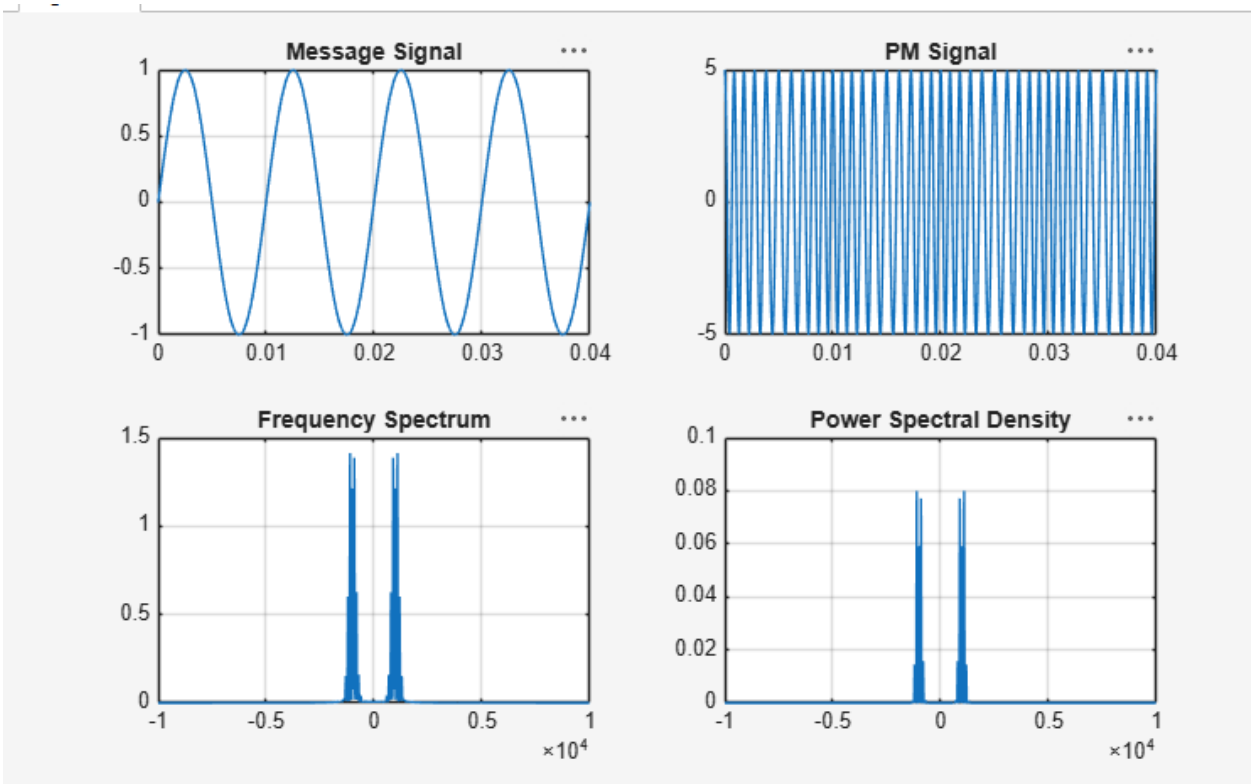
m=Am*sin(2*pi*fm*t);
pm=Ac*cos(2*pi*fc*t+kp*m);
beta=kp*Am;

N=length(pm);
f=(-N/2:N/2-1)*fs/N;
F=abs(fftshift(fft(pm)))/N;
PSD=F.^2*N/fs;
BW=2*(beta+1)*fm;

fprintf('Phase Deviation = %.2f radians\n',beta);
fprintf('Estimated Bandwidth = %.2f Hz\n',BW);

figure;
subplot(2,2,1); plot(t,m); title('Message Signal'); grid on;
subplot(2,2,2); plot(t,pm); title('PM Signal'); grid on;
subplot(2,2,3); plot(f,F); title('Frequency Spectrum'); grid on;
subplot(2,2,4); plot(f,PSD); title('Power Spectral Density'); grid on;
```

OUTPUT:



RESULT:

The Power Spectral Density (PSD) of the Phase Modulated (PM) signal was successfully obtained and analyzed using MATLAB.